

Programmazione e Algoritmi

12/10/2022



INSERTION SORT (A)

FOR $j = 2$ TO n DO

{ $k = A[j]$;

$i = j - 1$;

WHILE ($i > 0$ $\&\&$ $A[i] > k$)

{ $A[i+1] = A[i]$;

$i--$;
 $A[i+1] = k$;

$\forall j = 2 \dots n$, alla fine dell'iterazione j ^{esima} del FOR

• $A[1] \leq A[2] \leq \dots \leq A[j]$

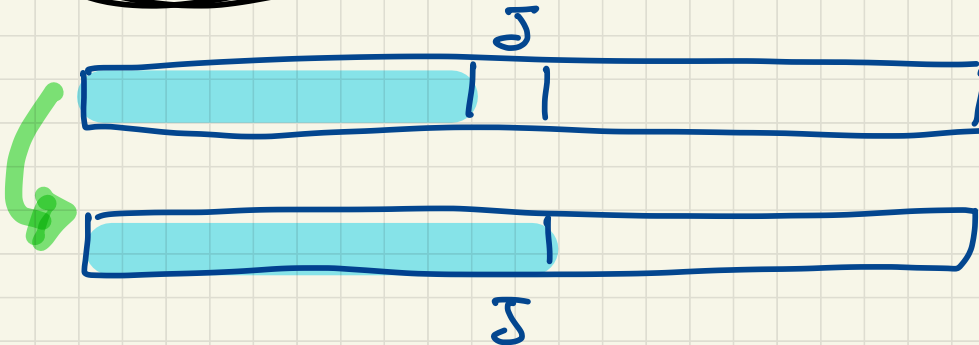
• $A[1] \dots A[j]$ sono i primi j elementi di A iniziale

INVARIANTE d.

CICLO:

IMPLICA LA CORRETTEZZA DI IS

$\forall j = 2 \dots n$



PRIMA

DOPO

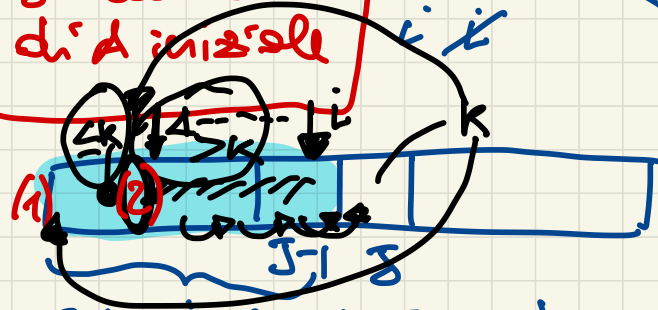
DIMOSTRO L'INVARIANTE DI CICLO PER INDUZIONE SU J

$\forall j = 2 \dots n$, alla fine dell'iterazione j^{esima} del FOR

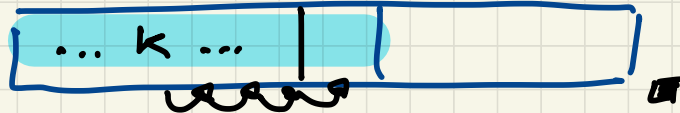
- $A[1] \leq A[2] \leq \dots \leq A[j]$
- $A[1] \dots A[j]$ sono i primi j elementi di A in ordine

- $A[1] \leq A[2] \leq \dots \leq A[3]$
- $A[1] \dots A[3]$ sono i primi 3 elementi di A in ordine

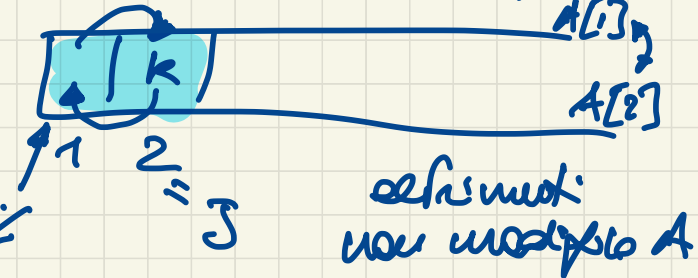
e) PASSO INDUTIVO
PRIMA



Sono i primi $J-1$ elementi di A iniziale
sono i primi J di A iniziale



1) USO BASE

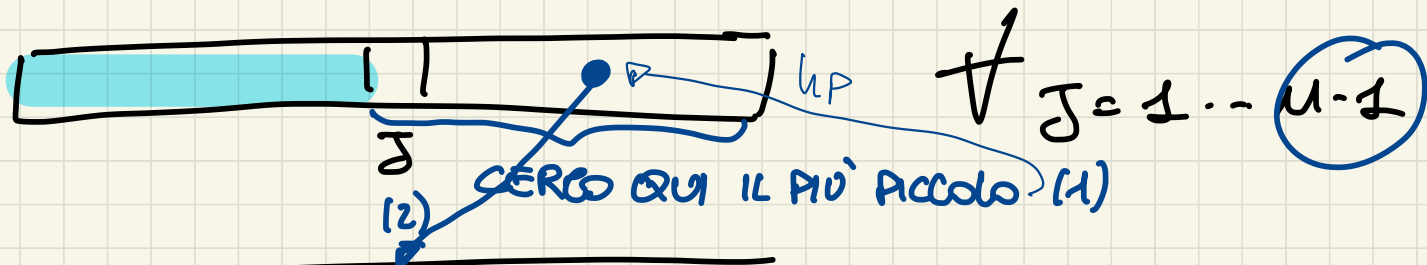
$$J=2$$


PER LP
(INDUTTIVA)

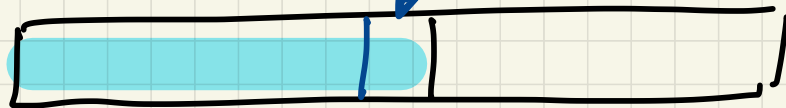
SELECTION SORT

È UN ALTRO ALGORITMO DI ORDINAMENTO ITERATIVO

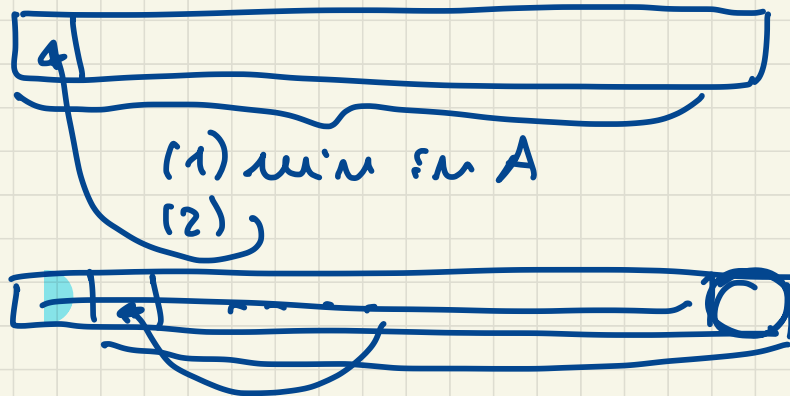
PRIMA



DOPO



INIZIO



SELECTION-SORT(A) #VOLTE EXECUTE

FOR $i = 1$ TO $n-1$ $\theta(i)$ $n-1$

{ $\text{minimo} = i$ $\theta(i)$ $n-1$

FOR $j = i+1$ TO n $\theta(i)$ $(n-1)(n-1) \dots \text{minimo}$

{ IF $A[j] < A[\text{minimo}]$ $\theta(i)$

THEN $\text{minimo} = j$; $\theta(i)$

SWAP $\{ A[i], A[\text{minimo}] \}$ $\theta(1)$
}

quanto volte?

46	42	4	15	3
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 A

3	42	4	15	46
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 $i=1$
 $\text{minimo} = 1$
 $i \rightarrow j \rightarrow 3 \rightarrow 4 \rightarrow 5$

3	4	42	15	46
---	---	----	----	----

 $i=2$
 $\text{minimo} = 2$
 $i \rightarrow j \rightarrow 3 \rightarrow 4 \rightarrow 5$

3	4	42	15	46
---	---	----	----	----

 $i=3$
 $\text{minimo} = 3$
 $i \rightarrow j \rightarrow 4 \rightarrow 5$

3	4	15	42	46
---	---	----	----	----

 $i=4$
 $\text{minimo} = 4$
 $i \rightarrow j \rightarrow 5$

3	4	15	42	46
---	---	----	----	----

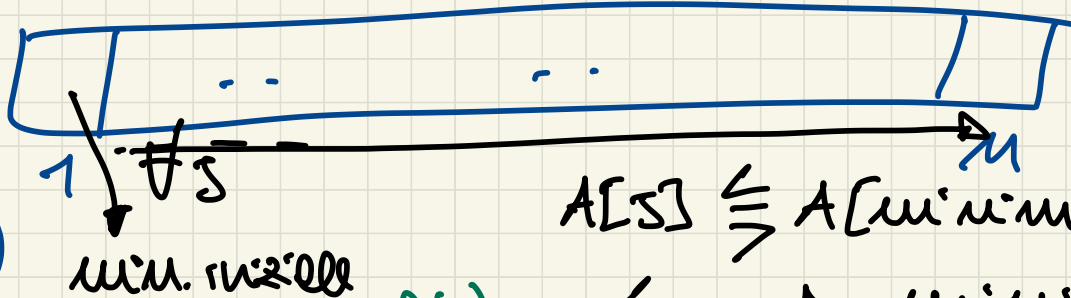
3	4	15	42	46
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MASSIMO (E MINIMO) IN ARRAY

Input : Array A di n elementi

Output : IL VALORE MINIMO in A

COMPLESSITA'
 $\Theta(n)$ AL CASO
OTTIMO
PESSIMO
MEDIO



MIN(A)

min. indice

minimo = 1 ; $\rightarrow \Theta(1)$

FOR $j = 2$ TO n DO $(n-1)$ volte $\geq \emptyset$

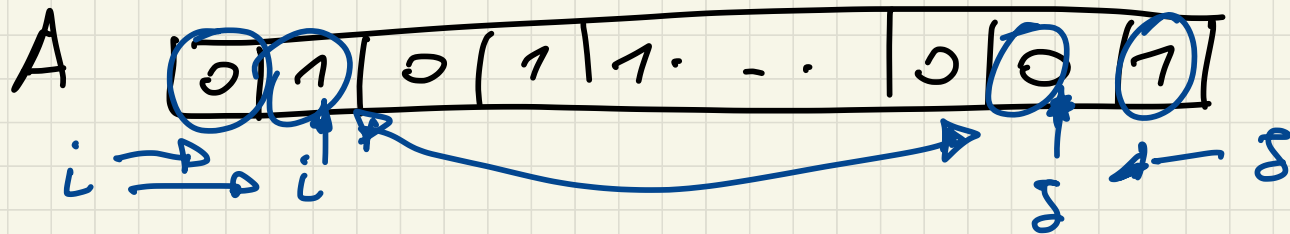
IF $A[j] < A[\text{minimo}]$ THEN minimo = j ; $\Theta(1)$

RETURN A[minimo]; $\Theta(1)$

ORDINAMENTO ARRAY BINARIO

Input A di n elementi in $\{0, 1\}$

Output A ordinato



A_{ord.} [0 0 0 0 . 0 | 1 1 - - - 1]

- 1) Conto gli 0 (o gli 1) e scivo l'A finale
2 SANZIONI LINEARI $\Rightarrow O(n)$
- 2) $\rightarrow$ CODICE SEQUE

ORDINA $\Theta(1)$ (A)

$i = 1;$

$j = n;$

• WHILE ($i < j$) DO $\leq n$ VOCTE

{ WHILE ($A[i] == 0$) DO i^{++} ;

WHILE ($A[j] == 1$) DO j^{--} ;

IF ($i < j$) THEN { SWAP($A[i], A[j]$);

i^{++} ;

j^{--} ;

}

— $O(n)$

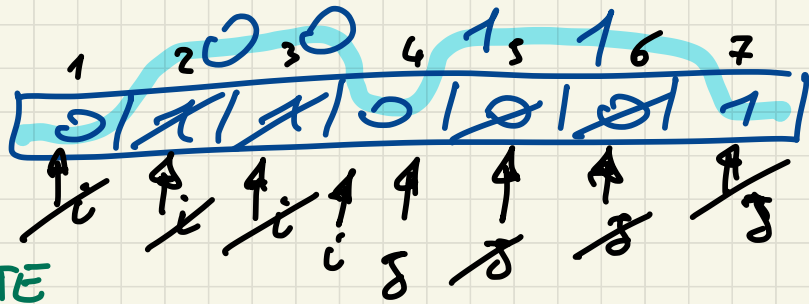
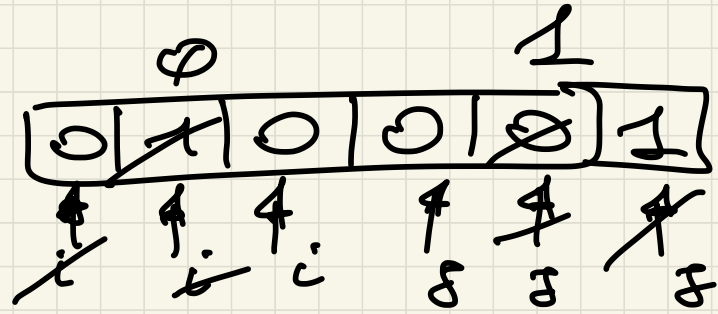
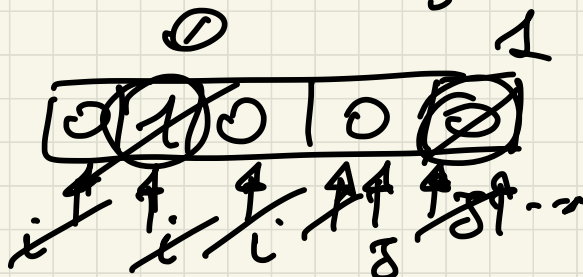
QUADRATIC?

— $O(n)$

no!

LINEAR

ORDINA
"IN-LOC"



CERCA i : $i = A[i]$

Input Dato un array A di n interi

Output i : $A[i] == i$ (se tale i esiste restituire -1 altrimenti)

A ORDINATO
interi sono
DISTINTI e
POSITIVI

(1)

3	7	5	4	9	10
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 $\rightarrow$ 4

1 2 3 4 5 6

3	7	5	5	10	9
---	---	---	---	----	---

 $\rightarrow$ -1